

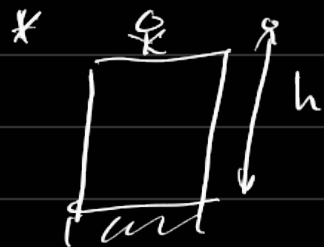
2.1

$$m \ddot{x}_{SE} + \gamma_{SE} \dot{x}_{SE} + k x_{SE} = 0$$

Mass

Wave number $\sim \frac{1}{\text{length}}$

For a spring: k : spring constant



$$F = ma = -mg$$

grav

a : constant

$$a = g$$

$$\frac{d^2 z}{dt^2} = \text{const}$$

$$z(t) = z_0 + t v_0 - \frac{1}{2} t^2 g$$

Defined: dimensionless parameters

$$\tau = \frac{t g}{h}$$

$$x_{SE} \rightarrow x$$

making our variables "dimensionless"

$$[x_{SE}] \sim [\text{meters}]$$

$$m \ddot{x}_{SE} + \gamma_{SE} \dot{x}_{SE} + k x_{SE} = 0$$

$$\ddot{x}_{SE} + \left(\frac{\gamma_{SE}}{m} \dot{x}_{SE} \right) + \left(\frac{k}{m} x_{SE} \right) = 0$$

$$\gamma = a \frac{\gamma_{SE}}{m}$$

$$\dot{x}_{SE} \sim \frac{dx}{dt} \sim \left(\frac{\text{meters}}{\text{sec}} \right)$$

$$\left[\frac{dx}{dt^2} \right] \sim \frac{m}{\text{sec}^2}$$

$$\left[\frac{\gamma_{SE}}{m} \dot{x}_{SE} \right] \sim \frac{m}{\text{sec}^2}$$

$$[m] = \text{kg}$$

$$\left[\frac{k}{m} \right] = \frac{1}{\text{sec}^2}$$

$$F_{\text{spring}} = -kx$$

(Hooke's Law)

$$\left[\frac{k}{m} \right] = \frac{1}{\text{sec}^2}$$

$$\left(\frac{\text{kg}}{\text{s}^2} \right) = (k?) (m)$$

$$[k] = \frac{\text{kg}}{(\text{sec})^2}$$

$$\left[\frac{\gamma_{SE}}{m} \dot{x}_{SE} \right] \frac{m}{\text{sec}^2}$$

$$\frac{k}{m} x_{SE} = x$$

$$\ddot{x} + \gamma \dot{x} + x = 0 \quad (157)$$

$$(156)$$

$$(eq) (156S)$$

$$T_0 = \sqrt{\frac{m}{k}}$$

$$m = 0.2 \text{ kg}$$

$$k = 0.5 \frac{\text{kg}}{\text{s}^2}$$

$\gamma = 0.8 \frac{\text{kg}}{\text{s}^2}$

$$\gamma = \frac{1}{2} \frac{r_{sr}}{\sqrt{mk}} = \frac{1}{2} \frac{(0.5 \frac{\text{kg}}{\text{s}^2})}{\sqrt{(0.2 \text{kg})(0.8 \frac{\text{kg}}{\text{s}^2})}} = \underline{\underline{0.625}}$$

* 3 cases :

- (i) Overdamped
- (ii) Underdamped
- (iii) Critically damped

} different values of γ

$\gamma = 1$: Critically damped

$\gamma > 1$: Overdamped

$\gamma < 1$: Underdamped



$$\ddot{x} + 2\gamma \dot{x} + x = 0$$

$$2(0.625) = 1.25$$

$$* \ddot{x} + 1.25 \dot{x} + x = 0$$

(eg) (150)

$$\lambda \pm i\omega = -\frac{\gamma_{sr}}{2m} \pm i \sqrt{\frac{k}{m} - \frac{\gamma_{sr}^2}{4m^2}}$$

$$\left| \gamma_{sr} \right|$$

(eg) (168)

$$\lambda = -\frac{5}{8} \pm i \sqrt{1}$$

$$\lambda = \pm i\omega - \frac{\gamma_{sr}}{2m} \pm i \sqrt{\frac{k}{m} - \frac{\gamma_{sr}^2}{4m^2}}$$

$$* \gamma_{sr} = 0.5 \text{ kg/s}$$

$$\omega = \sqrt{k/m} =$$

$$x = A e^{\lambda t} \quad (\text{Assume})$$

$$\ddot{x} = A \lambda^2 e^{\lambda t} = \lambda^2 x$$

$$\dot{x} = A \lambda e^{\lambda t} = \lambda x$$

$$\ddot{x} + 2\gamma \dot{x} + x = 0$$

$$(\lambda^2 + 2\gamma \lambda + 1) x = 0$$

$$(\lambda^2 + 2\gamma \lambda + 1) = 0$$

$$b = 2\gamma$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2} = -\gamma \pm \frac{\sqrt{4\gamma^2 - 4}}{2}$$

$$\lambda = -\gamma \pm i\sqrt{1-\gamma^2}$$

$$r = 0.625 = \frac{5}{8}$$

¶ In 7.2 they are solving for the ST ODE. Here we're solving for the simplest ODE

$$\lambda_{1,2} = -\frac{5}{9} \pm i \sqrt{1 - \frac{25}{64}} = -\frac{5}{9} \pm i \frac{\sqrt{39}}{8}$$

$$X(t) = A e^{\lambda_1 t} + B e^{\lambda_2 t}$$

$$\lambda_1 = -5/8 + i\sqrt{39}/8$$

$$\lambda_2 = -5/8 - i\sqrt{3}/8$$

$$x(t) = A e^{-s/8 t} \cdot e^{i \frac{\sqrt{3}}{8} t} + B e^{-s/8 t} e^{-i \frac{\sqrt{3}}{8} t}$$

$$x(t) = e^{-s/8t} \left[A e^{i\frac{\sqrt{30}}{8}t} + B e^{-i\frac{\sqrt{30}}{8}t} \right] \quad *$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta) \quad \text{vs: } \theta = \frac{\sqrt{39}}{8} t$$

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$$

$$= \cos(\theta) - i \sinh(\theta)$$

$$\left(A \left(\cos\left(\frac{\sqrt{3}t}{8}\right) + i \sin\left(\frac{\sqrt{3}t}{8}\right) \right) + B \left(\cos\left(\frac{\sqrt{3}t}{8}\right) - i \sin\left(\frac{\sqrt{3}t}{8}\right) \right) \right)$$

$$= \cos\left(\frac{\sqrt{30}e}{8}\right) \underbrace{(A+B)}_a + \sin\left(\frac{\sqrt{30}e}{8}\right) \underbrace{i(A-B)}_b$$

$$x(t) = e^{-5/8 t} \left(a \cos\left(\frac{\sqrt{39} t}{8}\right) + b \sin\left(\frac{\sqrt{39} t}{8}\right) \right) \quad (168)$$

- Next section adds a f^4 to RHS

$$M_{SE} \ddot{X}_{SE} + \gamma_{SE} \dot{X}_{SE} + X_{SE} = f(t)$$

